

WI.2 Permit Application Checklist

What to gather, in the order Wisconsin makes you gather it — and every code edition actually in force as of September 2026.

Before you use this document

This is a process reference, not legal advice. Wisconsin writes one Uniform Dwelling Code that binds every new one- and two-family dwelling in the state, and then splits the *enforcement* of it three ways — a municipality that inspects with its own staff, a municipality that contracts the work out, and a town that has never taken it up, where the state's own certified agency does it instead. Your septic permit comes from the county no matter which of the three you are in. Confirm every item with the office that will actually handle your parcel before you rely on it. Code editions change, and one of them changed this month. Confirm the electrical edition your permit falls under before you buy devices.

THE ORDER, AND WHY IT IS NOT THE ORDER YOU EXPECT

On a sewered lot, the building permit is the first thing you apply for. On an unsewered lot — which is most rural Wisconsin — it is close to the *last*. Two provisions put the county septic approval in front of your building permit, and neither is discretionary.

The sequence on an unsewered parcel

	Step	The part people miss
1	Soil and site evaluation by a certified soil tester	Three soil profile evaluations minimum, at least one dug as a full soil pit. You may not do this yourself
2	POWTS design and plans , sealed by a registered designer or signed by the master plumber who will install it	s. SPS 383.22(2)(c) — again, no owner-prepared option
3	County sanitary permit	The county has 30 days to decide once it has all the information and fees
4	Zoning / land use, shoreland, floodplain, driveway access	Local prerequisites; the building permit cannot issue until they are complete
5	Wisconsin uniform building permit — with erosion control	10 business days to approve or deny once the file is complete
6	Well — drilled any time before you need water; no routine state permit	But the separation distances in WI.4 constrain where the house and septic can go, so lay all three out together first

The two sentences that set the order

From the statute. *"No county, city, town or village may issue a building permit for construction of any structure requiring connection to a private on-site wastewater treatment system unless a private on-site wastewater treatment system satisfying all applicable regulations already exists to serve the proposed structure **or all permits necessary to install a private on-site wastewater treatment system have been obtained.**"* (Wis. Stat. s. 145.195(1))

From the code, on the building side. *"Pursuant to s. 145.195, Stats., if the proposed construction requires connection to a private onsite wastewater treatment system, a Wisconsin uniform building permit may not be issued unless conformance with s. SPS 383.25(2) has first been determined."* (s. SPS 320.09(9)(c))

The practical consequence. The soil tester is the critical path on a rural Wisconsin build, not the building inspector — and soil work has a season. POWTS components may not be installed *"if the soil is frozen at or below the infiltrative surface"*, snow cover must be removed before excavating, and if the soil *"can be rolled into a ¼-inch wire, the installation may not proceed"* (s. SPS 383.45(2), (3), (4)). Book the soil tester early in the year.

THE SANITARY PERMIT PACKAGE

- ☐ **Certified soil tester engaged.** A soil evaluation *"shall be performed by an individual who is a certified soil tester"* (s. SPS 385.10(1)), and since 1974 no person may even construct soil bore holes without that certificate (Wis. Stat. s. 145.045(1))
- ☐ **At least three soil profile evaluations**, and for a house — well under 1,000 gallons per day — *"at least one soil profile evaluation excavation per treatment or dispersal site shall be constructed as a soil pit"* (s. SPS 385.20(2)(b)1.a. and b.). A soil boring *"may not be created by means of a power auger"* (s. SPS 385.20(3)(b)2.)
- ☐ **Soil and site evaluation report** on the department's format, every page signed with the soil tester's identification number and the date (s. SPS 385.40(2))
- ☐ **POWTS plans** either sealed by a registered architect, engineer, designer of plumbing systems or designer of POWTS, *or* signed with license number by the master plumber responsible for installing it (s. SPS 383.22(2)(c))
- ☐ **Master Plumber or Master Plumber-Restricted Service named** — the application must carry *"documentation that the master plumber or the master plumber-restricted service who is to be responsible for the installation"* (s. SPS 383.21(2)(c)4.)
- ☐ **Sanitary permit application** filed with the county — *"the appropriate governmental unit where the POWTS is located or will be located"* (s. SPS 383.21(2)(b)1.). The permit runs to **you**, the property owner, not to the installer (Wis. Stat. s. 145.19(1g))
- ☐ **\$25 groundwater fee** — this one is statewide and fixed by statute (Wis. Stat. s. 145.19(6)). The base permit fee is set by your county and is not printed here
- ☐ **Management plan** submitted with the design — it is part of the plan submittal, and it sets your servicing intervals (s. SPS 383.54(1)(b))

Term and transfer: a sanitary permit “is valid for 2 years from the date of issue and renewable for similar periods” and “shall remain valid to the end of the established period, notwithstanding any change in the state plumbing code” (Wis. Stat. s. 145.19(7)) — so an issued permit locks its rules, though a renewal is judged against current rules (s. SPS 383.21(6)(b)). It may be transferred to a subsequent owner of the land, who must obtain a new copy from the issuing agent (s. 145.19(8)). Determination deadline: 30 days after the county has all required information and fees (s. SPS 383.21(3)(c)).

THE BUILDING PERMIT PACKAGE

Wisconsin's plan submittal is unusually short, and one sentence in it is worth real money to an owner-builder.

- ☐ **Wisconsin uniform building permit application**, on the form from the department, the municipality or your UDC inspection agency. *“No application shall be accepted that does not contain all the information requested on the form” (s. SPS 320.09(1))*
- ☐ **At least two sets of plans** (s. SPS 320.09(4)), *“legible and drawn to scale or dimensioned” (s. SPS 320.09(5))*
- ☐ **Site plan** showing the three items listed in the next section
- ☐ **The name of the initial downstream receiving water of the state** from the dwelling, for erosion and sediment control purposes (s. SPS 320.09(6)(d)) — look it up before you sit down to fill the form
- ☐ **Erosion and sediment control measures** shown on the site plan to comply with s. SPS 321.125
- ☐ **Energy compliance** — either the prescriptive table or a UA calculation; if you use REScheck it must be *“a version approved by the department” (s. SPS 322.31(2)(b))*
- ☐ **Sanitary permit already issued**, if you are not on a sewer
- ☐ **Every other local prerequisite complete** — the permit *“shall not be issued ... prior to the receipt of all completed forms, fees, plans, and documents required to process the application and completion of other local prerequisite permitting requirements” (s. SPS 320.09(2)(a)4.)*
- ☐ **The cautionary statement signed** (Wis. Stat. s. 101.65(1r); see WI.1)

Three rights on the building permit worth knowing

1. Nobody can make you buy a stamp. “Except as required under s. SPS 321.33, a municipality exercising jurisdiction may not require plans or calculations to be stamped or sealed by an architect or engineer.” (s. SPS 320.09(6)(c)) The one exception is floodplain construction — and even there, only specific items need a seal, notably the certification of the actual elevation of the lowest structural member by *“a registered land surveyor, architect or engineer” (s. SPS 321.33(3))*. On an ordinary house on ordinary ground, your own drawings are enough.

2. There is a shot clock. *“Action to approve or deny a uniform building permit application shall be completed within **10 business days** of receipt of all forms, fees, plans and documents required to process the application, and completion of other local prerequisite permitting requirements.”* (s. SPS 320.09(11))

3. You can start the hole before the full permit. *“Construction may begin on footings and foundations prior to the issuance of the Wisconsin uniform building permit where a permit to start construction is obtained.”* It takes a plot plan, complete footing and foundation information including exterior grading, and a fee — and *“the issuance of a permit to start construction shall not influence the approval or denial of the Wisconsin uniform building permit application.”* (s. SPS 320.09(9)(b)) In a short Wisconsin build season this is worth weeks.

Three more things happen at issuance. The **state seal** is affixed to the posted permit or the application, with the seal number on both, and the permit must be *“posted in a conspicuous place at the dwelling site”* (s. SPS 320.09(9)(d)). The seal costs **\$30.00** per new dwelling, paid to the department by whoever issues your permit (s. SPS 302.34(1)) — your municipality may bill a slightly higher line to cover handling. And *“the name and license number of the Wisconsin master plumber responsible for the installation of plumbing shall be entered on the permit by the issuing entity at the time of issuance”* (s. SPS 320.09(9)(a)7.).

The clock that runs against you

*“The permit shall expire **24 months** after issuance if the dwelling exterior has not been completed.”* (s. SPS 320.09(9)(a)5.) Note the trigger is the *exterior*, not the whole house — get the shell closed in. Two years is generous for a full-time build and tight for a weekends-and-evenings one, which is what most owner-builders are actually doing.

Building permit package: Wis. Admin. Code ss. SPS 320.08(1), 320.09(1), (2)(a)4., (4), (5), (6)(c), (6)(d), (9)(a)5., (9)(a)7., (9)(b), (9)(c), (9)(d), (11), 302.34(1), 321.33(3), 322.31(2)(b). Chapter SPS 320 as published, Register August 2026 No. 848.

WHAT THE SITE PLAN MUST SHOW — THE COMPLETE LIST

This is the entire statewide requirement. Three items, quoted in full.

Wis. Admin. Code s. SPS 320.09(5)(a)

“(a) Site plan. *The site plan shall show all of the following:”*

*“1. The location of the dwelling and any other buildings, **wells**, surface waters and **dispersal systems** on the site with respect to **property lines** and surface waters adjacent to the site.”*

*“2. The areas of **land-disturbing construction activity** and the location of all **erosion and sediment control measures** to be employed in order to comply with s. SPS 321.125.”*

“3. The *pre-construction ground surface slope and direction of runoff flow* within the proposed areas of land disturbance.”

Two things are notable. The well and the septic dispersal system are already required on the building-permit site plan — so draw all three together from the start. And there is **no specified scale**: “to scale or dimensioned” is the whole requirement. Anyone who tells you Wisconsin demands 1 inch = 100 feet on this drawing is adding a rule the code does not contain.

The sanitary-permit site plan is a different document with a different rule — see WI.4. Unlike this one, it may not be drawn by the owner: s. SPS 385.10(2) restricts site evaluation, including determining “land slope or setback distances to topographic or other site features”, to a closed list of credentialed people that does not include the property owner.

WHAT CODE IS ACTUALLY IN FORCE

The Uniform Dwelling Code is **Wisconsin's own code**, not an adoption of the International Residential Code. It is six chapters, and two of them are pointers to other chapters — which is why guides so often name the wrong one.

The six UDC chapters (“the code” = chs. SPS 320 to 325, s. SPS 320.07(16))

Chapter	What it covers	As published
SPS 320	Administration — permits, plans, inspections, adopted standards	Register August 2026 No. 848
SPS 321	Construction — structure, frost, loads, stairs, egress, floodplains	Register November 2024 No. 827
SPS 322	Energy conservation — Wisconsin's own prescriptive tables	Register November 2024 No. 827
SPS 323	Heating, ventilating and air conditioning	—
SPS 324	Electrical — a pointer , to ch. SPS 316	—
SPS 325	Plumbing — a pointer , to chs. SPS 381 to 387	—

THE ELECTRICAL CODE CHANGED ON 1 SEPTEMBER 2026

Almost every Wisconsin guide in circulation says the state is on the **2017** National Electrical Code. That was true until the first of this month. It is not true now.

*“The following standard is incorporated by reference into this chapter ... NFPA 70 National Electrical Code, (NEC) - **2023**, including all Temporary Interim Amendments and Errata prior to January 1, 2025.” (s. SPS 316.007(1)(a))*

The amendment is **CR 26-016, Register June 2026 No. 846, effective 1 September 2026**. Six NEC editions of change arrived at once — the 2017 book to the 2023 book — so expect differences in receptacle placement, GFCI and AFCI scope, service disconnects and surge protection. Do not wire from a 2017-vintage cheat sheet.

Which edition applies to you is a timing question. For additions and alterations the chapter is explicit: they *“shall comply with all provisions of this chapter at the time of permit application or, if no permit is required, the beginning of the project”* (s. SPS 316.003(6)). For new construction the chapter applies to *“all new installations”* (s. SPS 316.003(1)). If your permit application went in before 1 September, **ask your electrical inspector in writing which edition your job is being held to** and write the answer on this page.

Two related points people get wrong. First, the **UDC itself does not adopt the NEC** — NFPA 70 appears nowhere in the adoption tables at s. SPS 320.24. Electrical reaches your house through ch. SPS 316, and local electrical ordinances must *“strictly conform”* to the state code (Wis. Stat. s. 101.86(1)(a)). Second, **SPS 322 is not an IECC adoption**. Wisconsin writes its own envelope table; the only IECC reference on the page is a note that *“the IECC 2009 version of REScheck meets the thermal envelope requirements of this code.”* Saying Wisconsin “adopts the 2009 IECC” overstates what the chapter does.

Standards the UDC adopts by reference (s. SPS 320.24, Tables 320.24-1 to 320.24-13)

Standard	What it governs
SEI/ASCE 32-01	Design and Construction of Frost-Protected Shallow Foundations — the alternative to a 48-inch footing
ANSI/AWC NDS-2015	National Design Specification for Wood Construction
ANSI/AWC PWF-2007	Permanent Wood Foundation Design Specification
ANSI/TPI 1-2007	National Design Standard for Metal Plate Connected Wood Truss Construction
NFPA 54 / ANSI Z223.1 2015	National Fuel Gas Code — Wisconsin uses this rather than the IFGC
NFPA 13D 2013	Sprinklers in one- and two-family dwellings — adopted as a <i>standard</i> ; the UDC does not require sprinklers in a house
ICC 400-2012	Design and Construction of Log Structures — if you are building a log home, this is your standard
ACI 318-14, 332-14	Structural and residential concrete

*Selected from the adoption tables as published, Register August 2026 No. 848; the full tables also cover ASTM materials, ASHRAE handbooks, SMACNA duct standards, NAIMA and NIST references. An alternate standard “equivalent to or more stringent than” an adopted one may be used with department approval, and the department has **40 business days** to determine such an application (s. SPS 320.24(3)).*

THE TWO ZONE MAPS THAT RUN IN OPPOSITE DIRECTIONS

Wisconsin publishes two “Zone 1 / Zone 2” maps in the Uniform Dwelling Code. One sets your roof load. The other sets your insulation. **They are numbered the opposite way round.** Carry a zone number from one table to the other and you get both answers wrong, in opposite directions — under-framing a northern roof while under-insulating a northern wall.

Same words, opposite meanings

	Roof / snow load — Figure 321.02	Energy — s. SPS 322.31(1)(b)
Zone 1	the NORTH — roof load 40 psf	everything else — the southern and central counties
Zone 2	the SOUTH — roof load 30 psf	the 15 northern counties named below

Figure 321.02, “Zone Map for Roof Loads”, prints the legend “Zone 1 — 40 P.S.F. / Zone 2 — 30 P.S.F.” with Zone 1 covering the northern part of the state. Section SPS 322.31(1)(b) reads: “In Tables 322.31-1 and 322.31-2, zone 2 consists of the following 15 northern counties: Ashland, Bayfield, Burnett, Douglas, Florence, Forest, Iron, Langlade, Lincoln, Oneida, Price, Sawyer, Taylor, Vilas and Washburn. Zone 1 consists of all other counties not included in zone 2.” Both chapters as published, Register November 2024 No. 827.

Worked example, so this sticks

A house in **Vilas County** is roof-load **Zone 1 (40 psf)** and energy **Zone 2 (R-21 walls, R-38 floors)**. A house in **Dane County** is roof-load **Zone 2 (30 psf)** and energy **Zone 1 (R-20 walls, R-30 floors)**. The zone number flips; the severity does not. North is always the harder number in both tables — it is only the *label* that reverses.

Do not read a snow load off the map for your county from this kit. The zone boundary is drawn across county lines and there is a separate dashed line near the Door peninsula. Have your inspector or your truss supplier confirm the zone for the parcel, and write it on your permit record.

THE ENERGY TABLE, IN FULL

Table 322.31-1 — Insulation and Fenestration Requirements by Component

Component	Zone 1 — most of the state	Zone 2 — the 15 northern counties
Fenestration U-factor	0.35	0.35
Skylight U-factor	0.60	0.60
Ceiling R-value	49	49
Wood frame wall R-value	20 or 13+5	21
Mass wall R-value	15/19	19/21

Table 322.31-1 — Insulation and Fenestration Requirements by Component

Component	Zone 1 — most of the state	Zone 2 — the 15 northern counties
Floor R-value	30	38
Basement wall R-value	15/19	15/19
Crawl space wall R-value	10/13	10/13
Heated slab R-value	10/15	10/15
Unheated slab R-value	10	10

R-values are minimums, U-factors are maximums. “15/19” means R-15 continuous insulated sheathing on the interior or exterior of the home or R-19 cavity insulation at the interior of the basement wall, and may also be met with R-13 cavity plus R-5 continuous. “10/13” follows the same pattern. “13+5” means R-13 cavity plus R-5 insulated sheathing. R-20 and R-21 “may be compressed into a 2X6 cavity”. The floor value may instead be “insulation sufficient to fill the framing cavity with a minimum of R-19”. Heated-slab insulation must extend downward 48 inches, or to the bottom of the slab and then horizontally for a total of 48 inches. Table and footnotes as published, Register November 2024 No. 827.

There is a second way to comply. If the whole-dwelling envelope UA is at or below the UA you would get from the equivalent U-factors in Table 322.31-2 over the same areas, you comply — calculated by a method “consistent with the ASHRAE Handbook of Fundamentals” and including “the thermal bridging effects of framing materials” (s. SPS 322.31(2)(a)). That is the trade-off path REScheck automates.

Table 322.31-3 — Warm Air Furnaces and Boilers, Minimum Efficiency

Equipment type	Minimum efficiency
Natural gas and propane furnace	90% AFUE
Natural gas and propane hot water boilers	90% AFUE
Oil-fired furnaces	83% AFUE
Oil-fired hot water boilers	84% AFUE

Test procedures: DOE 10 CFR Part 430 or ANSI Z21.47 for gas and propane furnaces, DOE 10 CFR Part 430 for gas and propane hot water boilers and oil-fired hot water boilers, DOE 10 CFR Part 430 or UL 727 for oil-fired furnaces. There is a trade-off: in new construction a unit meeting only the federal standard may be installed “if the dwelling thermal envelope requirements of Table 322.31-4 are met” (s. SPS 322.31(3)(b)). Check that table with your inspector before counting on it — it is not reproduced here.

STRUCTURAL MINIMUMS YOU WILL DESIGN TO

Frost depth — s. SPS 321.16

*“Footings and foundations, **including those for landings and stoops**, shall be placed below the frost penetration level **or at least 48 inches below adjacent grade, whichever is deeper**, except as allowed under sub. (2).” and “Footings may not be placed on frozen material.” (s. SPS 321.16(1))*

Read the “whichever is deeper” carefully — **48 inches is a floor, not a target**. In the north the measured frost penetration governs and it is deeper. And note that landings and stoops are expressly caught, which is where owner-builders most often get written up.

Three exceptions exist (s. SPS 321.16(2)): a frost-protected shallow foundation designed to **SEI/ASCE 32-01**; the portion of a footing directly under a window areaway; and bearing **directly on bedrock** less than 48 inches down, provided the rock is cleaned of all earth, clay in crevices is removed to the frost penetration level or 1.5 times the crevice width, whichever is less, and water is prevented from collecting along the foundation.

Table 321.02-1 — Live Load (pounds per square foot)

Component	Live load (psf)
Floors	40
Garage floors	50
Exterior balconies, decks, porches	40
Ceilings (with storage)	20
Ceilings (without storage)	5

Wind: a dwelling shall withstand “either a horizontal and uplift pressure of 20 pounds per square foot acting over the surface area or the wind loads determined in accordance with ASCE 7-05” (s. SPS 321.02(1)(c)), with a Note that ASCE 7-05 “allows for substantial reduction from 20 psf”. Snow: roofs “shall be designed and constructed to support the minimum snow loads listed on the zone map” (s. SPS 321.02(1)(b)2.) — see the zone warning above.

EROSION CONTROL

Erosion control is not a separate adventure in Wisconsin — it rides on the building permit, and the same authority that issues one handles the other. What it does have is a specific slope trigger written into the statute: the rules “*shall require the use of more restrictive or additional practices on an area with a slope that is greater than **12 percent***” (Wis. Stat. s. 101.653(2)(b)). Your site plan has to show the pre-construction slope anyway, so measure it early.

☐

Land-disturbing areas and control measures shown on the site plan (s. SPS 320.09(5)(a)2.)

- ☐ Pre-construction ground surface slope and direction of runoff flow shown (s. SPS 320.09(5)(a)3.) — and check whether any of it exceeds 12 percent
- ☐ The initial downstream receiving water of the state identified (s. SPS 320.09(6)(d))
- ☐ If you will disturb **one acre or more**, ask your county or the Department of Natural Resources about a construction site stormwater permit — that is a separate DNR program, not part of your building permit
- ☐ Ask whether your municipality has imposed erosion requirements more stringent than the code, which s. SPS 320.02(2)(e)1. permits when directed by a federal EPA order or a DNR rule under s. NR 151.004

PERMIT RECORD — FILL THIS IN AS EACH ONE ISSUES

Every approval this build needs

	Item	Date	Notes
☐	Soil and site evaluation — certified soil tester Tester / cert. no. _____ Report date _____		
☐	County sanitary permit — POWTS Permit no. _____ Expires _____		
☐	Master plumber engaged for the POWTS Name _____ License no. _____		
☐	Zoning / land use permit Office _____ Permit no. _____		
☐	Shoreland or floodplain approval, if applicable Office _____ Ref. _____		
☐	Driveway / access permit Authority _____ Permit no. _____		
☐	Address assigned (rural 911 addressing) Address _____		
☐	Wisconsin uniform building permit Permit no. _____ Seal no. _____		
☐	Enforcing authority / UDC inspection agency Name _____		

Every approval this build needs

	Item	Date	Notes
☐	Erosion control approval Ref. _____ Slope over 12%? _____		
☐	Electrical code edition my permit is held to 2017 or 2023 NEC _____ Confirmed by _____		
☐	Roof-load zone and energy zone for this parcel Roof zone _____ Energy zone _____		
☐	Well construction report filed by the driller Driller _____ Date _____		
☐	Water sample results — coliform and nitrate Date _____ Result _____		

Sources — every Wisconsin claim in this document (verified September 2026)

What this document states	Authority
Sanitary permit before the building permit	Wis. Stat. s. 145.195(1); s. SPS 320.09(9)(c)
\$25 groundwater fee; 2-year term; transferable	Wis. Stat. s. 145.19(6), (7), (8)
County has 30 days to decide the sanitary permit	s. SPS 383.21(3)(c)
Soil evaluation by a certified soil tester; three evaluations, one a pit	ss. SPS 385.10(1), 385.20(2)(b); Wis. Stat. s. 145.045(1)
Two sets of plans; the three-item site plan	s. SPS 320.09(4), (5)(a)
No architect or engineer seal may be required	s. SPS 320.09(6)(c); exception s. SPS 321.33
Ten business days to approve or deny	s. SPS 320.09(11)
Footing and foundation permit to start construction	s. SPS 320.09(9)(b)
Permit expires at 24 months if the exterior is not complete	s. SPS 320.09(9)(a)5.
\$30 state seal per new dwelling	s. SPS 302.34(1)
2023 National Electrical Code, effective 1 September 2026	s. SPS 316.007(1)(a); CR 26-016, Register June 2026 No. 846

Sources — every Wisconsin claim in this document (verified September 2026)

What this document states	Authority
The UDC does not itself adopt the NEC	Negative finding — Tables 320.24-1 to 320.24-13
Roof-load zones 40 and 30 psf; energy zones inverted	Figure 321.02; s. SPS 322.31(1)(b)
The prescriptive envelope table and its footnotes	Table 322.31-1
90% AFUE gas furnaces and boilers	Table 322.31-3
48 inches or below frost penetration, whichever is deeper	s. SPS 321.16(1)(a)
Live loads; 20 psf wind alternative	Table 321.02-1; s. SPS 321.02(1)(c)
Erosion control: more restrictive above 12 percent slope	Wis. Stat. s. 101.653(2)(b)

Everything cited in this kit is free to read at **docs.legis.wisconsin.gov**, and the site takes a citation directly as a URL. For a statute — a **Wis. Stat.** section such as 101.65 — open **docs.legis.wisconsin.gov/statutes/statutes/101** and find the section in the chapter. For a rule — a **Wis. Admin. Code** chapter such as SPS 321 or NR 812 — open **docs.legis.wisconsin.gov/code/admin_code** and pick the agency: **SPS** is Safety and Professional Services, which writes the Uniform Dwelling Code and the septic code, and **NR** is Natural Resources, which writes the well code and shoreland zoning. **One habit worth keeping when you check a number:** every chapter page prints its register date at the foot, in the form “Register month year No. nnn”. That date, not the year in the chapter title, tells you how current the text on your screen is. All text here was read at that source, September 2026.